BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, MARDAN

INTERMEDIATE (12TH CLASS) EXAMINATION, 2023

MATHEMATICS

Past Paper 2023 • Paper Code: PR – XII (01) (23)

Retyped faithfully from the original board paper — same questions, no changes.

SECTION-A

Time Allowed: 20 Min | **Maximum Marks:** 20 | **Paper Code:** PR – XII (01) (23)

Q1. SECTION-A

Q1. Domain of function f(x) = |x| is____.

(A) $(0,\infty)$ (B) $(-\infty,\infty)$

(C) $(-\infty,0)$ (D) $(0,1)$

Q2. The function f(x) = √(x−3) is continuous for all x and____.

(A) $x>3$ (B) $x\leq 3$

(C) $x<3$ (D) $x\geq 3$

Q3. lim $(e^x-1)/x$ = ____. (x→0)

(A) 0 (B) e

(C) 1 (D) x

Q4. d/dx (cos x) = ____.

(A) sin x (B) -sin x

(C) tan x (D) sec x

Q5. If $f(x) = 2x^3 - 3x^2 + 4$ then f'(2) = ____.

(A) 10 (B) 12

(C) 14 (D) 8

Q6. lim tanx/x = ____. (x→0)

(A) 0 (B) 2

(C) 1 (D) 3

Q7. IF y = sin x then y''' = ____.

(A) sin x
(B) -sin x
(C) cos x
(D) -cos x

Q8. The graph of a function f(x) is concave downward on (a,b) where:

(A) f''(x)>0
(B) f''(x)<0
(C) f''(x)=0
(D) None of these

Q9. A function f(x) is decreasing if____.

(A) f'(x)>0
(B) f'(x)<0
(C) f'(x)=0
(D) f''(x)=0

Q10. $\int_0^1 x^2 dx$ = ____.

(A) 0
(B) −1/3
(C) 2/3
(D) 1/3

Q11. ∫cotx dx = ____.

(A) ln|cos x|+c
(B) ln|sin x|+c
(C) tan x+c
(D) ln|tan x|+c

Q12. The slope of the line y−1=0 is____.

(A) -1
(B) 0
(C) 1
(D) None of these

Q13. The distance between two points A(3,-2) and B(-1,-5) is____.

(A) 10
(B) 11
(C) 5
(D) 4

Q14. The center of circle $x^2+y^2+2gx+2fy+c=0$ is____.

(A) (g,f)
(B) (-g,-f)
(C) (g,-f)
(D) (-g,f)

Q15. The parabola $y^2=4px$, if p<0 then it is open____.

(A) Down
(B) Up
(C) Left
(D) Right

Q16. If the eccentricity e<1, the conic is called____.

(A) Ellipse
(B) Parabola
(C) Circle
(D) Hyperbola

Q17. $y^2/25 - x^2/4 = 1$ is an equation of____.

(A) Circle
(B) Parabola
(C) Ellipse
(D) Hyperbola

Q18. Order of the differential equation $((d^3y/dx^3)^2 - (d^2y/dx^2)^4 + dy/dx + y - 0$ is____.

(A) 4
(B) 3
(C) 2
(D) 1

Q19. f(x,y) = (√x + √y)/(x+y) is a homogenous function of degree____.

(A) −1/2
(B) 1/2
(C) 0
(D) 1

Q20. $\int_0^1 e^{x^2}$ dx has____.

(A) One actual solution
(B) More than one actual solutions
(C) No actual solution
(D) Both A and B

SECTION-B

> Note: Time allowed for Section-B and Section-C is 2 Hours and 40 minutes.

Q2. Answer any TEN parts. Each part carries FIVE marks.

1) Find the inverse function of f(x) = (3x+1)/5.

2) Evaluate lim (tanθ−sinθ)/$\sin^3\theta$ (θ→0)

3) Use first principal rule to determine the derivative of $f(x) = 5x^2 - 3x$?

4) Find dy/dx by using any suitable rule of differentiation $y = x^2e^{\sin(x)}$

5) If f(x) = ln(4x+7), find f^{iv}(x).

6) Evaluate lim [sin(3t)/3t î + ln(2+t) ĵ] (x→0)

7) Find the Maclaurin series expansion for the function $f(x)=\sin^2 x$.

8) Evaluate $\int x^3 \cos x\, dx$

9) Use the integration by parts toe valuate $\int x^2 \cos x\, dx$

10) Find the centroid of the triangle ABC whose vertices are A(3,5), B(4,6) and C(3,-1).

11) Find the equation of parabola with vertex (-3,4) and passes through (5,11) and its line of symmetry is horizontal.

12) If $f(x,y)=x^2y+xy^2$, then find f_x and f_y.

13) Solve the initial value problem $dy/dx = 1/x^2$, y(2)=0

SECTION-C

Note: Attempt any THREE questions. All questions carry equal marks.

Q3.

(a) Find dy/dx if $y = ((2x-5)/(x-4))^4$.

(b) Find the angle θ from the line 2x+4y-10=0 to the line 5x-3y+1=0

Q4.

(a) Find the equation of tangent to the circle $x^2+y^2=9$ which makes an angle of 60° with x-axis?

(b) Find the critical value of $f(x) = x^2/(x-2)$

Q5.

(a) Find the ∫_{π/2}^{π} cos(x/2 + π) dx

(b) Find an equation of a circle which passes through the points (1,2), (3,-4) and (5,-6).

Q6.

(a) Solve the differential equation x dy/dx = x + y

(b) Find the equation of hyperbola with vertices are at (5,0) and (-5,0) and one focus at (-7,0).